

ERS

RADIANODE ENGINEERING READINESS SERIES

ERS-2026-001

Engineering Readiness Assessment

INT8 Post-Training Quantization for Seq2Point NILM on Resource-Constrained Edge Platforms

INDEPENDENT ENGINEERING VALIDATION

PUBLICATION CLASSIFICATION

FIELD	VALUE
Reference	ERS-2026-001
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Author	Oluwanifemi Adelana Ogunjemilua
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Reproducibility	Supported
Supporting artefacts	github.com/radianode/ers-2026-001

EXECUTIVE SUMMARY

This Engineering Readiness Assessment evaluates whether INT8 post-training quantization provides a deployment-ready optimisation for a Seq2Point Non-Intrusive Load Monitoring (NILM) model intended for resource-constrained Edge AI platforms.

The investigation was conducted using the Radianode engineering validation methodology, focusing on the engineering characteristics that determine deployment readiness rather than theoretical model performance.

The measured evidence demonstrates that the evaluated optimisation:

- Reduced deployment memory by 75%
- Preserved model behaviour
- Maintained prediction accuracy
- Introduced no observable deployment-critical numerical instability

Based on the engineering evidence presented in this assessment, the evaluated model satisfies the criteria for deployment readiness.

ENGINEERING DECISION

Can INT8 post-training quantization be adopted as a deployment optimisation for the evaluated Seq2Point NILM model without compromising engineering confidence?

This publication answers that question using measured engineering evidence.

INVESTIGATION SCOPE

ITEM	DESCRIPTION
Engineering asset	Seq2Point NILM model
Engineering objective	Evaluate whether INT8 post-training quantization preserves deployment readiness of a Seq2Point NILM model for resource-constrained Edge AI platforms
Optimisation evaluated	INT8 post-training quantization
Deployment scenario	Resource-constrained Edge AI
Evaluation dataset	UK-DALE
Model architecture	Seq2Point
Implementation framework	MATLAB Deep Learning Toolbox
RVP	RVP is Radianode's structured methodology for turning engineering evidence into defensible decisions. RVP connects the engineering question, expected behaviour, controlled evidence and resulting decision so that conclusions remain traceable to the evidence behind them.

ENGINEERING PROCESS**RVP** · Radianode Validation Process

RVP is Radianode's structured methodology for turning engineering evidence into defensible decisions. It connects the engineering question, expected behaviour, controlled evidence and resulting decision so that conclusions remain traceable to the evidence behind them.

STAGE	DEFINITION	APPLICATION IN ERS-2026-001
RVP-01 Assess	Define the engineering problem by reviewing requirements, understanding system architecture, identifying constraints, and establishing the scope of assessment.	Defined the engineering objective, deployment scenario, and evaluation scope for INT8 post-training quantization of a Seq2Point NILM model.
RVP-02 Simulate	Apply appropriate engineering analysis to evaluate expected system behaviour.	Performed engineering evaluation of the optimised model using representative residential energy data.
RVP-03 Validate	Verify engineering assumptions using objective evidence, performance evaluation, documented findings, and structured engineering review.	Assessed memory efficiency, behaviour preservation, prediction accuracy, and numerical stability through repeatable engineering workflows.
RVP-04 Decide	Evaluate the evidence against requirements, assumptions and remaining uncertainty to establish readiness.	Determined deployment readiness based on the measured engineering evidence presented in this assessment.

01

Memory Efficiency

ENGINEERING QUESTION

Has deployment memory been reduced sufficiently to provide a meaningful engineering benefit?

MODEL	DEPLOYMENT MEMORY
FP32	354.10 KB
INT8	88.52 KB
Measured reduction	75%

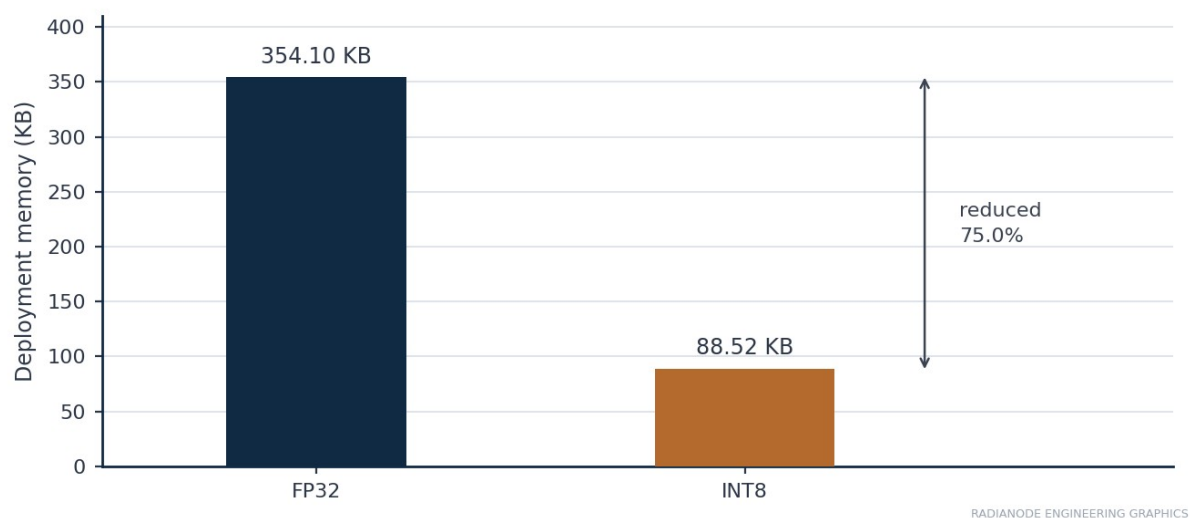


FIGURE 1
Memory Footprint Comparison

Deployment memory for the FP32 and INT8 models. Quantization reduced the footprint from 354.10 KB to 88.52 KB, a reduction of 75.0%.

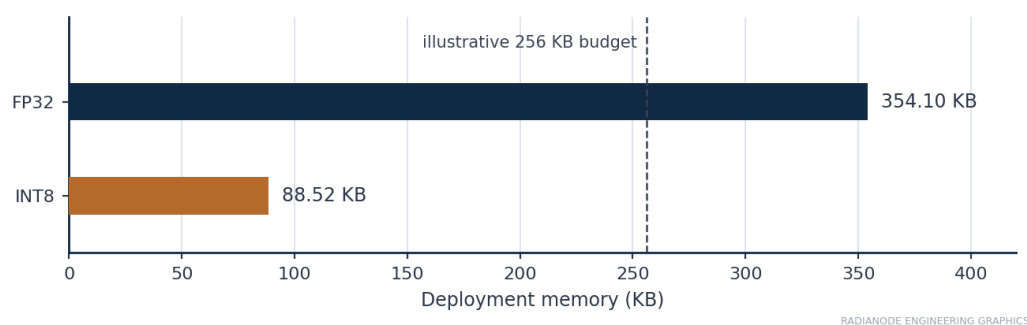


FIGURE 2
Model Memory Breakdown

The same two footprints on a linear scale, shown against an illustrative 256 KB platform budget. The budget line is illustrative and is not drawn from a named target platform.

ENGINEERING INTERPRETATION

The optimisation reduced deployment memory by 75% without modifying the network architecture or requiring model retraining. This represents a significant improvement in deployment efficiency and increases suitability for memory-constrained embedded platforms.

ENGINEERING FINDING

Memory efficiency improved with no observed engineering compromise.

Behaviour Preservation

ENGINEERING QUESTION

Did the optimised model continue behaving like the original implementation?

METRIC	RESULT
Correlation	0.9940
Mean difference	0.8532 W
Maximum difference	2.7062 W

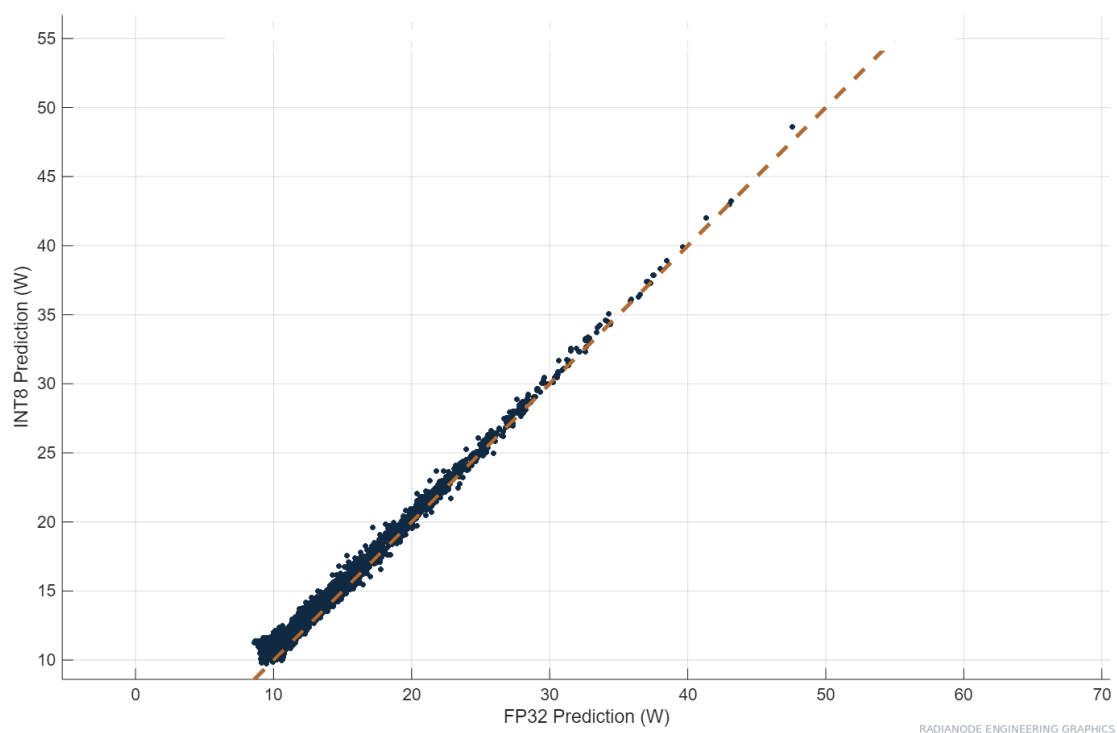


FIGURE 3
FP32 vs INT8 Prediction Scatter

Each point is one prediction pair. The copper line is the identity relation, not a fit. Dispersion about that line is the quantization error, and it does not widen with magnitude.

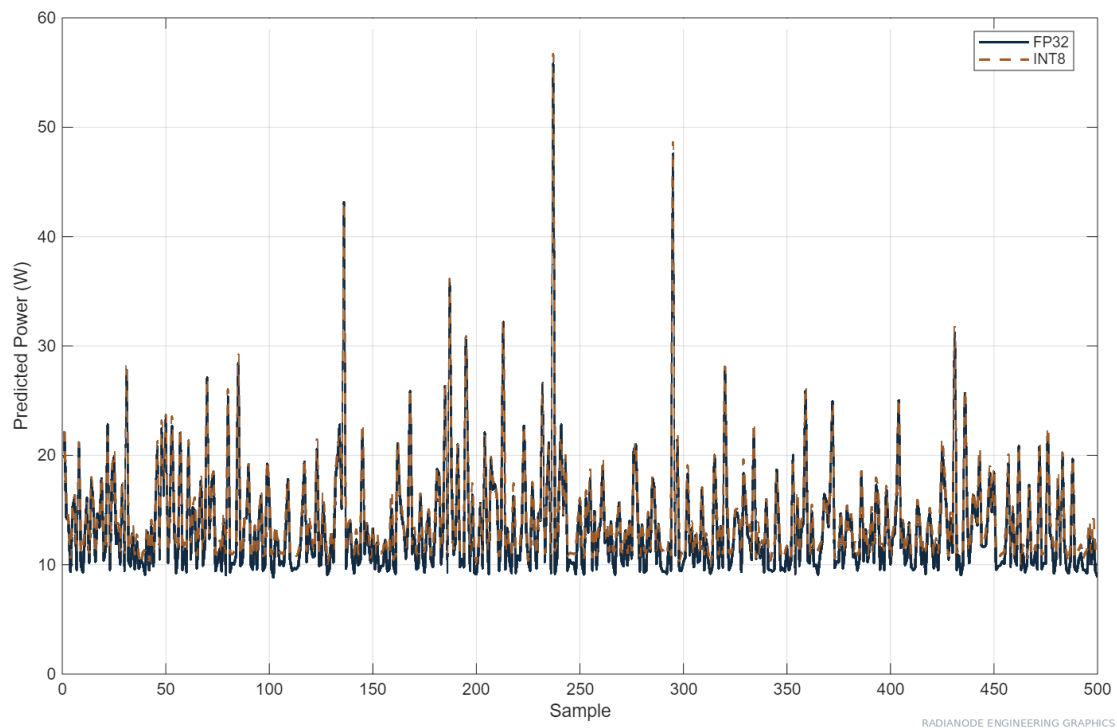


FIGURE 4
Prediction Overlay

FP32 and INT8 traces overlaid across the first 500 samples. The two follow the same transitions, including the isolated high-power events.

ENGINEERING INTERPRETATION

Prediction behaviour remained highly consistent following optimisation. The measured correlation demonstrates that the optimised model preserved the operational characteristics of the original implementation.

ENGINEERING FINDING

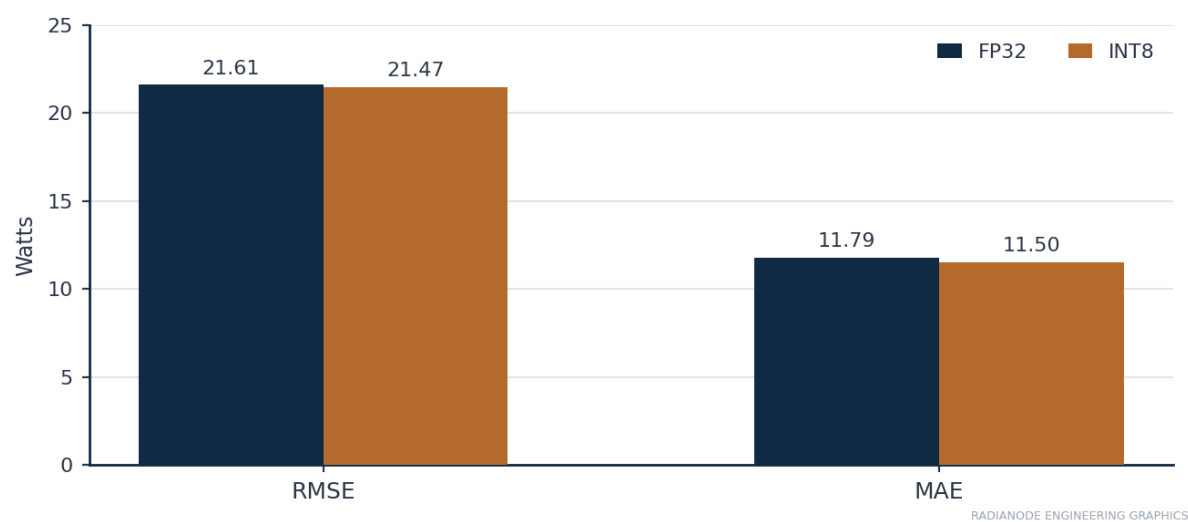
Model behaviour remained consistent after optimisation.

Prediction Accuracy

ENGINEERING QUESTION

Was inference performance maintained following optimisation?

METRIC	FP32	INT8
RMSE	21.6061 W	21.4706 W
MAE	11.7867 W	11.5046 W



RADIANODE ENGINEERING GRAPHICS

FIGURE 5
Accuracy Comparison

RMSE and MAE for both models against ground truth. The differences between the two are smaller than the spread that would be expected from retraining alone.

ENGINEERING INTERPRETATION

Prediction accuracy remained effectively unchanged. The measured differences are operationally insignificant and provide no evidence that optimisation reduced inference performance.



ENGINEERING FINDING

Prediction accuracy was maintained after optimisation.

Numerical Stability

ENGINEERING QUESTION

Did optimisation introduce deployment-critical numerical effects?

METRIC	RESULT
FP32 to INT8 RMSE	1.0254 W
FP32 to INT8 MAE	0.8532 W
Mean relative error	7.90%

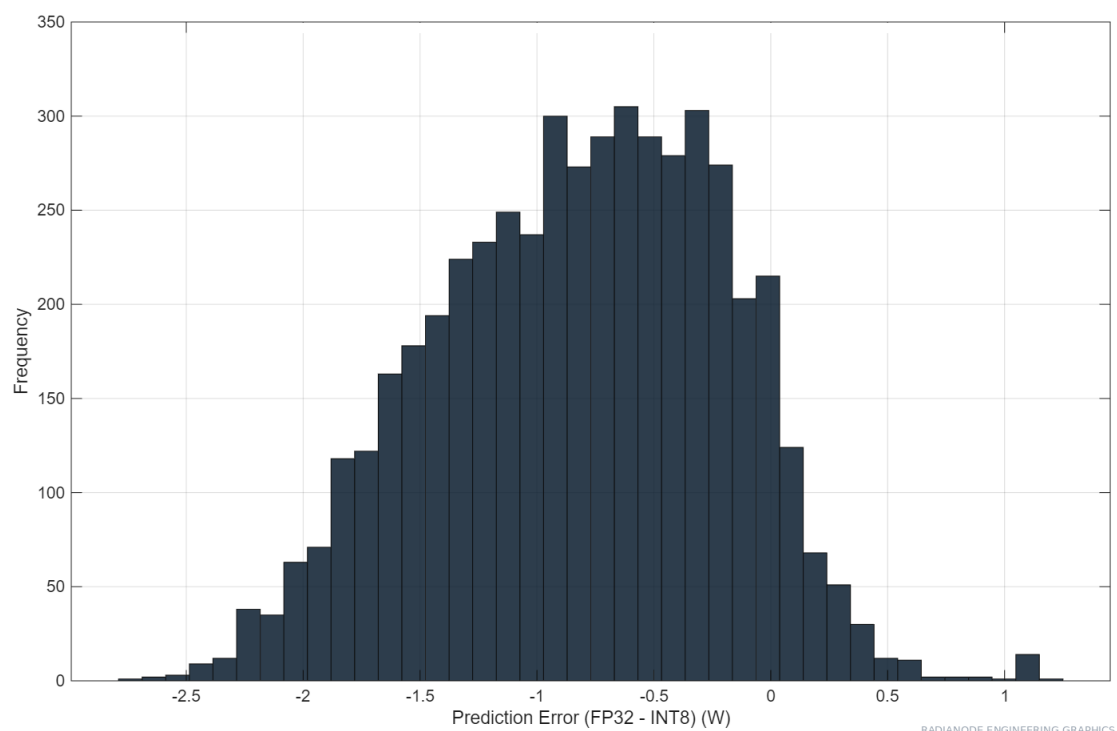


FIGURE 6
Error Distribution Histogram

Distribution of the prediction difference between the two models. The distribution is unimodal and bounded, with no secondary mode and no heavy tail.

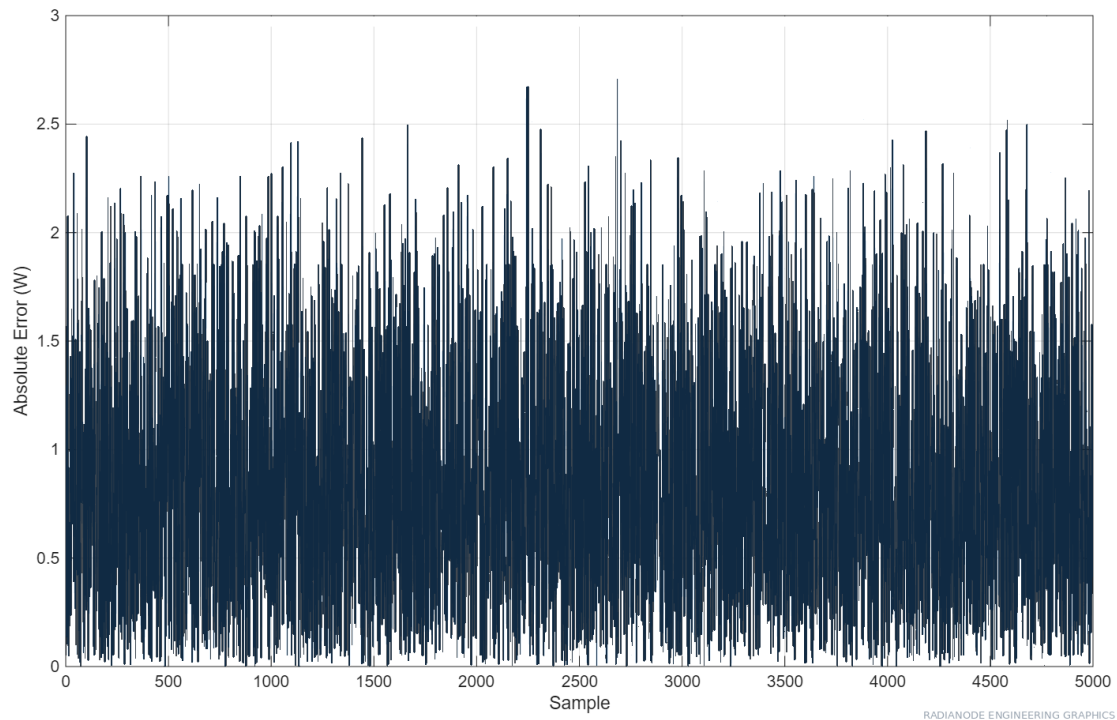


FIGURE 7
Absolute Error

Absolute prediction difference against sample index across 5,000 samples. The envelope is stable across the sequence, which is the evidence that the deviation is not accumulating.

ENGINEERING INTERPRETATION

The observed numerical differences remained small, predictable and non-systematic throughout the evaluation.

The measured relative error did not introduce observable changes in prediction behaviour or inference quality.

Within the scope of this engineering assessment, the measured numerical deviation provides no evidence of deployment-critical numerical instability.

ENGINEERING FINDING

Numerical stability remained within acceptable engineering limits.

SCOPE OF EVALUATION

Clearly defining the limits of this assessment strengthens the credibility of the engineering conclusions presented.

This assessment evaluates

- Deployment memory efficiency
- Behaviour preservation
- Prediction accuracy
- Numerical stability

This assessment does not evaluate

- Hardware execution latency
- Energy consumption
- Processor utilisation
- Hardware-specific optimisation
- Cross-dataset generalisation
- Hardware-in-the-loop deployment
- Long-term operational performance

ENGINEERING EVIDENCE SUMMARY

ENGINEERING CRITERION	OUTCOME
Memory efficiency	Confirmed
Behaviour preservation	Confirmed
Prediction accuracy	Maintained
Numerical stability	Verified

RADIANODE ENGINEERING POSITION

Following independent engineering evaluation, Radianode considers INT8 post-training quantization to be an appropriate deployment optimisation for the evaluated Seq2Point NILM model when deployment objectives prioritise memory efficiency without compromising inference performance.

For the evaluated deployment scenario:

- Deployment memory was significantly reduced.
- Prediction behaviour remained consistent.
- Inference accuracy was maintained.
- Numerical stability remained within acceptable engineering limits.

Based on the measured evidence, no engineering barriers to deployment were identified.

Deployment validation on the intended production hardware remains recommended as part of final system acceptance.

ENGINEERING READINESS OUTCOME

Ready for edge deployment.

REPRODUCIBILITY STATEMENT

This Engineering Readiness Assessment is an implementation of the Radianode Validation Process (RVP).

Supporting engineering artefacts, reproducibility resources, implementation workflows, and publication assets are available through the official Radianode GitHub repository.

Repository contents

- Engineering implementation
- Evaluation workflows
- Figure-generation scripts
- Engineering metrics
- Publication assets
- Reproducibility guide
- Version history

Repository

github.com/radianode/ers-2026-001

APPENDIX A

Figure Traceability

Every published figure is traceable to the engineering workflow that generated it.

REPORT FIGURE	ENGINEERING WORKFLOW	OUTPUT
Figure 1 – Memory Footprint Comparison	Memory evaluation	Memory comparison chart
Figure 2 – Model Memory Breakdown	Memory evaluation	Deployment memory chart
Figure 3 – FP32 vs INT8 Prediction Scatter	Behaviour comparison	Prediction fidelity scatter plot
Figure 4 – Prediction Overlay	Behaviour comparison	Prediction overlay plot
Figure 5 – Accuracy Comparison	Accuracy evaluation	RMSE/MAE comparison chart
Figure 6 – Error Distribution Histogram	Numerical stability evaluation	Error distribution histogram
Figure 7 – Absolute Prediction Error	Numerical stability evaluation	Absolute prediction error plot

This appendix strengthens reproducibility without exposing or depending on a specific implementation toolchain.

RADIANODE LTD

Independent Engineering Firm

This publication (ERS-2026-001) is issued by Radianode Ltd, an independent UK-registered engineering firm. Supporting evidence, raw data and reproducibility materials are available at the repository referenced in this report's Reproducibility Statement.

hello@radianode.com · radianode.com